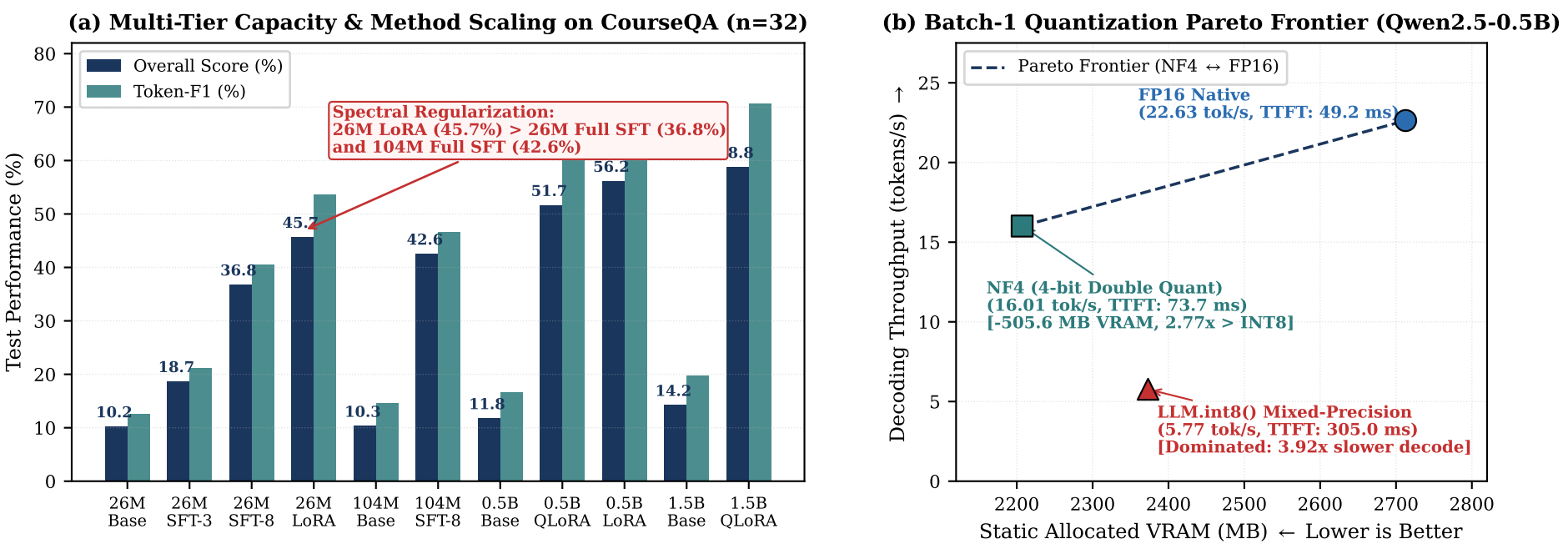
(a) Multi-Tier Capacity & Method Scaling on CourseQA (n=32)
Overall Score (%)
Token-F1 (%)
Spectral Regularization:
26M LoRA (45.7%) > 26M Full SFT (36.8%)
and 104M Full SFT (42.6%)
Test Performance (%)
10.2
18.7
36.8
45.7
10.3
42.6
11.8
51.7
56.2
14.2
58.8
26M Base
26M SFT-3
26M SFT-8
26M LoRA
104M Base
104M SFT-8
0.5B Base
0.5B QLoRA
0.5B LoRA
1.5B Base
1.5B QLoRA
(b) Batch-1 Quantization Pareto Frontier (Qwen2.5-0.5B)
Pareto Frontier (NF4 ↔ FP16)
FP16 Native
(22.63 tok/s, TTFT: 49.2 ms)
NF4 (4-bit Double Quant)
(16.01 tok/s, TTFT: 73.7 ms)
[-505.6 MB VRAM, 2.77x > INT8]
LLM.int8() Mixed-Precision
(5.77 tok/s, TTFT: 305.0 ms)
[Dominated: 3.92x slower decode]
Decoding Throughput (tokens/s) ↑
Static Allocated VRAM (MB) ← Lower is Better